

1. Maximum likelihood estimator of σ^2

Set the first derivative to zero:

$$\frac{\partial \ell(\sigma^2)}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{\sum_{t=1}^T (x_t - x_{t-1})^2}{2\sigma^4} = 0.$$

Multiply both sides by $2\sigma^4$:

$$-T\sigma^2 + \sum_{t=1}^T (x_t - x_{t-1})^2 = 0.$$

Solving for σ^2 gives the MLE:

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (x_t - x_{t-1})^2$$

This is the usual sample variance of the increments.

2. Second-order condition

Your second derivative is:

$$\frac{\partial^2 \ell(\sigma^2)}{\partial (\sigma^2)^2} = \frac{T}{2\sigma^4} - \frac{\sum_{t=1}^T (x_t - x_{t-1})^2}{(\sigma^2)^3}.$$

Evaluating it at $\hat{\sigma}^2$:

$$\left. \frac{\partial^2 \ell(\sigma^2)}{\partial (\sigma^2)^2} \right|_{\hat{\sigma}^2} = -\frac{T}{2(\hat{\sigma}^2)^2} < 0,$$

confirming that $\hat{\sigma}^2$ is a **maximum**.

3. Variance of the estimator (Fisher information)

The Fisher information for σ^2 is

$$\mathcal{I}(\sigma^2) = -\mathbb{E} \left[\frac{\partial^2 \ell(\sigma^2)}{\partial (\sigma^2)^2} \right].$$

Using $\mathbb{E}[(x_t - x_{t-1})^2] = \sigma^2$,

$$\mathcal{I}(\sigma^2) = \frac{T}{2(\sigma^2)^2}.$$

Hence, the asymptotic variance of the MLE is

$$\boxed{\text{Var}(\hat{\sigma}^2) = \frac{1}{\mathcal{I}(\sigma^2)} = \frac{2\sigma^4}{T}}$$

In practice, replace σ^2 with $\hat{\sigma}^2$:

$$\text{Var}(\hat{\sigma}^2) \approx \frac{2(\hat{\sigma}^2)^2}{T}.$$

4. Asymptotic distribution

By standard likelihood theory,

$$\sqrt{T} (\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}(0, 2\sigma^4),$$

or equivalently,

$$\hat{\sigma}^2 \sim \mathcal{N}\left(\sigma^2, \frac{2\sigma^4}{T}\right).$$

Summary (ready to quote)

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (x_t - x_{t-1})^2, \quad \text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{T}.$$